

Algebra: Evaluating Composition of Functions

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DIRECTIONS

Evaluate each composition. Substitute the inner function result into the outer function.

1 Find $f(g(2))$:

$$f(x)=2x+1, \quad g(x)=x+3$$

Answer: _____

2 Find $g(f(3))$:

$$f(x)=2x+1, \quad g(x)=x+3$$

Answer: _____

3 Find $f(g(0))$:

$$f(x)=x^2-1, \quad g(x)=x+2$$

Answer: _____

4 Find $g(f(1))$:

$$f(x)=x^2-1, \quad g(x)=x+2$$

Answer: _____

5 Find $f(g(-1))$:

$$f(x)=3x, \quad g(x)=x^2+1$$

Answer: _____

6 Find $g(f(-1))$:

$$f(x)=3x, \quad g(x)=x^2+1$$

Answer: _____

7 Find $f(g(4))$:

$$f(x)=\sqrt{x}, \quad g(x)=x-4$$

Answer: _____

8 Find $f(g(8))$:

$$f(x)=\sqrt{x}, \quad g(x)=x-4$$

Answer: _____

9 Find $(f \circ g)(2)$:

$$f(x)=x+5, \quad g(x)=2x$$

Answer: _____

10 Find $(g \circ f)(2)$:

$$f(x)=x+5, \quad g(x)=2x$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES

Work inside-out: evaluate $g(x)$ first, then substitute that value into f . Order matters: $f(g(x)) \neq g(f(x))$ in general.

1 $f(x)=2x+1$, $g(x)=x+3$; $f(g(2))$

= 11

$g(2)=5$, then $f(5)=2(5)+1=11$

2 $f(x)=2x+1$, $g(x)=x+3$; $g(f(3))$

= 10

$f(3)=7$, then $g(7)=7+3=10$

3 $f(x)=x^2-1$, $g(x)=x+2$; $f(g(0))$

= 3

$g(0)=2$, then $f(2)=4-1=3$

4 $f(x)=x^2-1$, $g(x)=x+2$; $g(f(1))$

= 2

$f(1)=0$, then $g(0)=0+2=2$

5 $f(x)=3x$, $g(x)=x^2+1$; $f(g(-1))$

= 6

$g(-1)=2$, then $f(2)=3(2)=6$

6 $f(x)=3x$, $g(x)=x^2+1$; $g(f(-1))$

= 10

$f(-1)=-3$, then $g(-3)=9+1=10$

7 $f(x)=\sqrt{x}$, $g(x)=x-4$; $f(g(4))$

= 0

$g(4)=0$, then $f(0)=\sqrt{0}=0$

8 $f(x)=\sqrt{x}$, $g(x)=x-4$; $f(g(8))$

= 2

$g(8)=4$, then $f(4)=\sqrt{4}=2$

9 $f(x)=x+5$, $g(x)=2x$; $(fog)(2)$

= 9

$g(2)=4$, then $f(4)=4+5=9$

10 $f(x)=x+5$, $g(x)=2x$; $(gof)(2)$

= 14

$f(2)=7$, then $g(7)=2(7)=14$



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